

MAT2007 Project Report

Data Analysis: Metabolic Flux Distributions

Research Question: How do eight different microbial organisms compare in their experimentally measured metabolic flux distributions across glycolysis, the TCA cycle, and the pentose phosphate pathway?

Asya Özzeybek (i6423633)

Date: October 15, 2026

Introduction

Metabolism, which is a fundamental feature of all cells, is composed of numerous interconnected reactions that transform metabolites [1]. Metabolic flux refers to the rate at which metabolites are converted through these reactions [2]. Studying metabolic fluxes holds promising prospects for quantifying microbial metabolism, understanding the mechanisms of diseases, testing drug responses, and improving bioprocess engineering. Glycolysis, the tricarboxylic acid (TCA) cycle, and the pentose phosphate pathway (PPP) are key pathways in central carbon metabolism, supporting energy production and providing precursors for biosynthesis in microbial organisms [3].

This study addresses the question: **"How do eight different microbial organisms compare in their experimentally measured metabolic flux distributions across glycolysis, the TCA cycle, and the pentose phosphate pathway?"** The data from the Central Carbon Metabolic Flux Database (CeCaFDB) was analyzed using Python; comparing the magnitude, variability, and shape of the flux distributions through descriptive statistics and boxplots. The descriptive statistics used involves mean, median, interquartile range, standard deviation (using N as the denominator), skewness, and excess kurtosis. The data was analyzed quantitatively while noting that the observed variations could arise from both distinct organismal metabolisms and differences in experimental conditions

Data Sample

The data set obtained from CeCaFDB contains flux values compiled from scientific articles involving different experimental conditions[1]. Flux values are reported without units, having been normalized to a substrate uptake rate of 100. The microbial organisms included in this analysis are: *Actinobacillus succinogenes*, *Arthrobacter* sp., *Ashbya gossypii*, *Aspergillus nidulans*, *Aspergillus niger*, *Bacillus megaterium*, *Bacillus subtilis*, and *Chlorobaculum tepidum*. The data cleaning process was required to remove rows with missing flux values and rows with KEGG reaction IDs that do not fall under the three pathways (see Table 1).

Table 1: Number of flux observations included in the analysis after cleaning and pathway filtering.

Species	**Glycolysis**	**TCA**	**PPP**	**Total**
Actinobacillus succinogenes	10	4	6	20
Arthrobacter sp.	30	30	24	84
Ashbya gossypii	20	20	12	52
Aspergillus nidulans	6	24	9	39
Aspergillus niger	20	20	12	52
Bacillus megaterium	70	63	42	175
Bacillus subtilis	9	9	6	24
Chlorobaculum tepidum	16	16	12	44
Total	**181**	**186**	**123**	**490**

Additionally, debugging and testing was implemented to ensure that missing flux values were removed, flux values were numeric, reaction IDs were correctly matched, and statistical calculations were aligned with the anticipated values.

Results and Discussion

The main results illustrated that *C. tepidum* had the highest central values in TCA with the highest mean of 120.7 and the highest median of 98.1, while *A. gossypii* showed the highest variability with a standard deviation of 136.733 in glycolysis. The clear differences between the central values and the spread of the distributions were observed both from the quantitative results and data visualization.

[1] http://www.cecafdb.org/ [4]

Figure 1 displays the general comparison of the flux distribution across the eight microbial organisms that combine the three pathways along with outliers. *A. nidulans* and *B. subtilis* have relatively high median flux values, while *A. gossypii* and *B. megaterium* show much wider distributions, indicating greater variability.

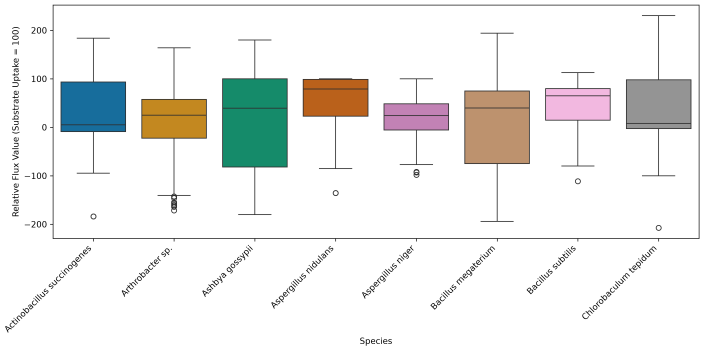


Figure 1: Normalized Flux Value Distributions Across the Eight Microbial Organisms

C. tepidum has the highest mean TCA flux of 120.7 but the lowest mean glycolytic flux of −0.55. The combination of pathways masks this contrast, making separate pathway boxplots more informative in terms of the research question (see Figure 2).

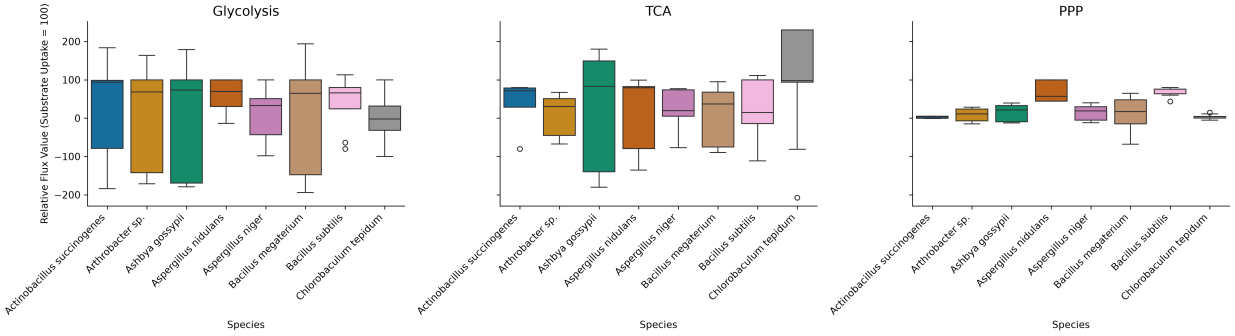


Figure 2: Normalized Flux Value Distributions of the Pathways of Glycolysis, TCA Cycle and PPP Across the Eight Microbial Organisms

It can be noticed that glycolysis and TCA generally show much wider distributions compared to PPP which looks to be more compact for most organisms. The eight microbial organisms showed species and pathway dependent flux distributions. *C. tepidum* had the highest mean TCA reaction flux but the lowest mean glycolysis reaction flux while *A. gossypii* showed particularly high variability in both pathways. Differences in experimental conditions and also in the number of reactions represented for each organism limit the comparison. Another limitation is that a mean flux near zero does not necessarily correspond to low pathway activity because positive and negative flux values across different reactions can cancel each other. Future implication could be comparing corresponding reactions under matched conditions to investigate whether these differences persist, and applying the Kruskal-Wallis test which was not applied, as the focus of this project is quantitative exploratory comparison rather than formal hypothesis testing.

AI disclosure: ChatGPT assisted with Python debugging, summarizing important findings, and LaTeX formatting.

References

[1] Whyte, B. (2026). *Metabolic flux: the dynamic flow of biochemical reactions.* https://www.bmglabtech.com/en/blog/metabolic-flux-the-dynamic-flow-of-biochemical-reactions/

[2] ScienceDirect Topics (n.d.). *Metabolic Flux.* https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/metabolic-flux

[3] Martinez, A., Dijkstra, P., Megonigal, P., and Hungate, B. A. (2024). *Microbial central carbon metabolism in a tidal freshwater marsh and an upland mixed conifer soil under oxic and anoxic conditions.* https://doi.org/10.1128/aem.00724-24

[4] CeCaFDB (n.d.). *Central Carbon Metabolic Flux Database.* http://www.cecafdb.org/